MATHEMATICS

TARGET : JEE-ADVANCED-2024

CAPS-11^{+}

SCQ (Single Correct Type) :

1. The minimum value of $x^2 + y^2 + z^2$ if $ax + by + cz = p$, is :

(A) $\left(\frac{p}{a+b+c}\right)^2$ (B) $\frac{p^2}{a^2+b^2+c^2}$ (C) $\frac{a^2+b^2+c^2}{p^2}$ (D) 0

2. The distance between the line $\vec{r} = 2\hat{i} - 2\hat{j} + 3\hat{k} + \lambda(\hat{i} - \hat{j} + 4\hat{k})$ and the plane $\vec{r}\cdot(\hat{i} + 5\hat{j} + \hat{k}) = 5$ is :

(A) $\frac{10}{9}$ (B) $\frac{10}{3\sqrt{3}}$ (C) $\frac{3}{10}$ (D) $\frac{10}{3}$

3. If tangents drawn from each point on the line $3x + 4y - 11 = 0$ upon a parabola are at right angle and chords joining points of contact on parabola for each point on line $3x + 4y - 11 = 0$ pass through a fixed point (2, 5) then length of latus rectum of parabola is equal to :

(A) 3 (B) 6 (C) 12 (D) 5

4. If $Ax + By = 1$ is a normal to the curve $ay = x^2$, then :

(A) $4A^2(1 - aB) = aB^3$ (B) $4A^2(2 + aB) = aB^3$

(C) $4A^2(1 + aB) + aB^3 = 0$ (D) $2A^2(2 - aB) = aB^3$

5. The equation of a curve which passes through the point (3, 1), such that the segment of any tangent between the point of tangency and the x–axis is bisected at its point of intersection with y-axis, is :

(A) $x = 3y^2$ (B) $x^2 = 9y$ (C) $x = y^2 + 2$ (D) $2x = 3y^2 + 3$

MCQ (One or more than one correct) :

6. $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 2$ and $\vec{a}\cdot\vec{b} = 1$, If $\vec{c} = (2\vec{a}\times\vec{b}) - 3\vec{b}$, then

(A) $|\vec{c}| = 2\sqrt{3}$ (B) $|\vec{c}| = 4\sqrt{3}$ (C) $\vec{b} \wedge \vec{c} = \frac{2\pi}{3}$ (D) $\vec{b} \wedge \vec{c} = \frac{5\pi}{6}$

7. Let $\vec{u}$ be a vector in the x – y plane with slope $\sqrt{3}$. Further $|\vec{u}|, |\vec{u} - \hat{i}|, |\vec{u} - 2\hat{i}|$ are in G.P. $\hat{i}$ being the unit vector along positive x-axis, then $|\vec{u}|$ is equal to

(A) $\sqrt{3 - 2\sqrt{2}}$ (B) $\sqrt{3 + 2\sqrt{2}}$ (C) $\tan\frac{9\pi}{8}$ (D) $\cot\frac{3\pi}{8}$

8. If $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-coplanar vectors, then $\left[\vec{a} \times (\vec{b} + \vec{c})\ \vec{b} \times (\vec{c} - 2\vec{a})\ \vec{c} \times (\vec{a} + 3\vec{b})\right]$ is equal to

(A) $\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2$ (B) $7\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2$

(C) $-5\left[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}\right]$ (D) $7\left[\vec{c}\times\vec{a}\ \ \vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\right]$

9. If θ is the angle between the vectors $\vec{p} = a\hat{i} + b\hat{j} + c\hat{k}$ and $\vec{q} = -b\hat{i} + c\hat{j} + a\hat{k}$, where a, b, c $\in$ R, then all possible values of θ lies in

(A) $\left[0, \frac{5\pi}{6}\right]$ (B) $\left[\frac{5\pi}{6}, \pi\right]$ (C) $\left[\frac{\pi}{3}, \frac{2\pi}{3}\right]$ (D) $\left[0, \frac{2\pi}{3}\right]$

10. Consider the lines :

$$L_1 : \frac{x-2}{1} = \frac{y-1}{7} = \frac{z+2}{-5}$$

$L_2 : x - 4 = y + 3 = -z$

Then which of the following is/are correct ?

(A) Point of intersection of L_1 and L_2 is (1, – 6, 3)

(B) Equation of plane containing L_1 and L_2 is x + 2y + 3z + 2 = 0

(C) Acute angle between L_1 and L_2 is $\cot^{-1}\left(\frac{13}{15}\right)$

(D) Equation of plane containing L_1 and L_2 is x + 2y + 2z + 3 = 0

11. The value(s) of μ for which the straight lines $\vec{r} = 3\hat{i} - 2\hat{j} - 4\hat{k} + \lambda_1(\hat{i} - \hat{j} + \mu\hat{k})$ and $\vec{r} = 5\hat{i} - 2\hat{j} + \hat{k} + \lambda_2(\hat{i} + \mu\hat{j} + 2\hat{k})$ are coplanar is/are :

(A) $\frac{5+\sqrt{33}}{4}$ (B) $\frac{-5+\sqrt{33}}{4}$ (C) $\frac{5-\sqrt{33}}{4}$ (D) $\frac{-5-\sqrt{33}}{4}$

12. The line with direction cosines proportional to 2, 1, 2 meets each of the lines x = y + 1 = z at P and x – 2y + 1 = 0 & y – z = 0 at Q. Then coordinates of each of the points of intersection are

(A) sum of coordinates of P is 8 (B) sum of coordinates of Q is 3

(C) PQ = 3 (D) PQ = 4

13. Given the equation of the line 3x – y + z + 1 = 0, 5x + y + 3z = 0. Then which of the following is (are) correct.

(A) symmetrical form of the equation of line is $\frac{x}{2} = \frac{y - \frac{1}{8}}{-1} = \frac{z + \frac{5}{8}}{1}$

(B) symmetrical form of the equation of line is $x = \frac{x + \frac{1}{8}}{1} = \frac{y - \frac{5}{8}}{1} = \frac{z}{-2}$

(C) equation of the plane through (2, 1, 4) and perpendicular to the given lines is 2x – y + z – 7 = 0

(D) equation of the plane through (2, 1, 4) and perpendicular to the given lines is x + y – 2z + 5 = 0

14. Two distinct tangents are drawn to parabola $y^2 = 4x$ from P(h, k) then (given h $\neq$ 0).

(A) If slopes of both the tangents are positive then hk > 0

(B) if h < 0, slopes of the two tangents are of different signs

(C) If product of slopes of tangents is negative and hk > 0, then sum of slopes is positive

(D) If product of slopes of tangents is negative and hk > 0, the sum of slopes is negative

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15. PQ is a double ordinate of the parabola $y^2 = 4ax$. If the normal at P intersect the line passing through Q and parallel to x-axis at G; then locus of G is a parabola with :

(A) Vertex at (4a, 0)
(B) focus at (5a, 0)
(C) Directrix as the line x = 3a – 0
(D) length of latus rectum equal to 4a

Numerical Based Questions :

16. If $\vec{a},\vec{b}$ are two unit vectors and $\vec{c}$ is such that $\vec{c} = \vec{a}\times\vec{c}+\vec{b}$, then the maximum value of $\left[\vec{a}\,\vec{c}\,\vec{b}\right]$ is $\frac{p}{q}$ where p and q are co-prime natural numbers, then find the value of (p + q)

17. If $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = 2$, and $\vec{d}$ is unit vector then find the value of $\left|\left(\vec{a}\times\vec{b}\right)\times\left(\vec{c}\times\vec{d}\right)+\left(\vec{b}\times\vec{c}\right)\times\left(\vec{a}\times\vec{d}\right)+\left(\vec{c}\times\vec{a}\right)\times\left(\vec{b}\times\vec{d}\right)\right|$.

18. Let $\vec{a}$ and $\vec{b}$ be any two unit vectors, then find the greatest positive integer in the range of the expression $\frac{3}{2}|\vec{a}+\vec{b}|+2|\vec{a}-\vec{b}|$

19. Points A and B lie on the parabola $y = 2x^2 + 4x - 2$, such that origin is the mid-point of the line segment AB. If 'l' be the length of the line segment AB, then find the unit digit of l^2.

Subjective Type Problems :

20. Let OABC be a tetrahedron whose edges are of unit length. If $\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = \vec{b}$ and $\overrightarrow{OC} = \alpha(\vec{a}+\vec{b})+\beta(\vec{a}\times\vec{b})$, then $(\alpha\beta)^2 = \frac{p}{q}$ where p and q are relatively prime to each other.

Find the value of $\left[\frac{q}{2p}\right]$ where [·] denotes greatest integer function.

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ANSWERKEY OF CAPS – 11+

1.	(B)	**2.**	(B)	**3.**	(B)	**4.**	(D)	**5.**	(A)	**6.**	(BD)	**7.**	(ACD)
8.	(BD)	**9.**	(AD)	**10.**	(ABC)	**11.**	(AC)	**12.**	(ABC)	**13.**	(BD)	**14.**	(ABC)
15.	(ABCD)	**16.**	3	**17.**	4	**18.**	5	**19.**	8				